

SCOPE OF WORK

VXI Hardware Operability, Unity Plan Wind Tunnel, NASA Langley Research Center

Contract Type: Firm Fixed Price

Period of Performance: To be determined after order acceptance.

1. BACKGROUND AND PURPOSE

This Scope of Work covers technical services to make the VXI (VMEbus eXtensions for Instrumentation) hardware operational for the Unity Plan Wind Tunnel facility at NASA Langley Research Center. The work involves reviewing and modifying the existing code base, updating drivers, and simplifying the application structure to improve operational efficiency and maintainability.

2. CONTRACTOR INFORMATION

Sole Source Justification: Original writers of code. No known source for repair of code.

3. NASA POINT OF CONTACT

Technical Representative:

Organization: NASA Langley Research Center

4. SCOPE OF WORK

The Contractor shall provide technical services to restore and optimize VXI hardware operations for the Unity Plan Wind Tunnel facility. Work shall be performed in three distinct milestones as detailed below.

5. DELIVERABLES AND MILESTONES

Milestone 1: KSC Driver Development for MV177

Tasks:

- Review the existing code base installed at the facility

- Build KSC driver for MV177 system
- Address unique driver requirements:
 - Implement hardware query capability
 - Configure memory-mapped AB Blue hose
 - Identify and configure base addresses on KSC VXI boards
 - Clarify and document card identification and addressing methodology

Deliverable: Functional KSC driver for MV177 with documentation

Milestone 2: Application Structure Modification

Tasks:

- Simplify the application architecture
- Modify directory structure to allocate one directory per Input/Output Controller (IOC)
- Enable boot capability from any server
- Test conversion of ADL files to Phoebe display format
- Consolidate records for each IOC into a single .xls file to enable mass editing capabilities

Deliverable: Simplified application structure with consolidated configuration files and conversion test results

Milestone 3: On-Site Installation, Testing, and Demonstration

Tasks:

- Travel to NASA Langley Research Center for on-site implementation
- Make VXI hardware operational
- Make AB Blue hose operational
- Create new application boot directory with updated database
- Boot IOCs from new directory structure

- Demonstrate modification process from .xls to .db file creation

Deliverable: Fully operational VXI system with demonstration of maintenance procedures

6. SCHEDULE

Period of performance is 90 days ARO. The Contractor commits to completing all three milestones within the agreed-upon timeframe.

7. PRICING AND PAYMENT TERMS

Contract Type: Firm Fixed Price

Payment Schedule:

- Milestone 1 completion
 - Milestone 2 completion
 - Milestone 3 completion
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8. PLACE OF PERFORMANCE

- **Remote Work:** Milestones 1 and 2 (Contractor's facility)
 - **On-Site Work:** Milestone 3 (NASA Langley Research Center, Hampton, Virginia)
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9. ACCEPTANCE CRITERIA

Each milestone shall be considered complete upon:

- Successful completion of all tasks listed under the milestone
 - Delivery of specified deliverables
 - Acceptance by NASA Technical Representative
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10. CONTRACTOR RESPONSIBILITIES

The Contractor shall:

- Provide all labor, materials, and expertise necessary to complete the work
 - Coordinate with NASA technical personnel for access and requirements
 - Provide progress updates as requested
 - Ensure all work meets professional industry standards
 - Maintain communication with the NASA point of contact
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11. NASA RESPONSIBILITIES

NASA shall:

- Provide access to existing code base and hardware systems
 - Provide facility access for on-site work (Milestone 3)
 - Designate a technical point of contact
 - Review and approve milestone completions
 - Process invoices according to payment terms
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12. ASSUMPTIONS AND CONSTRAINTS

- Contractor assumes existing hardware is functional and accessible
- NASA will provide necessary documentation and access credentials
- Any additional work beyond the three defined milestones will require a contract modification
- Travel costs for Milestone 3 are included in the milestone price